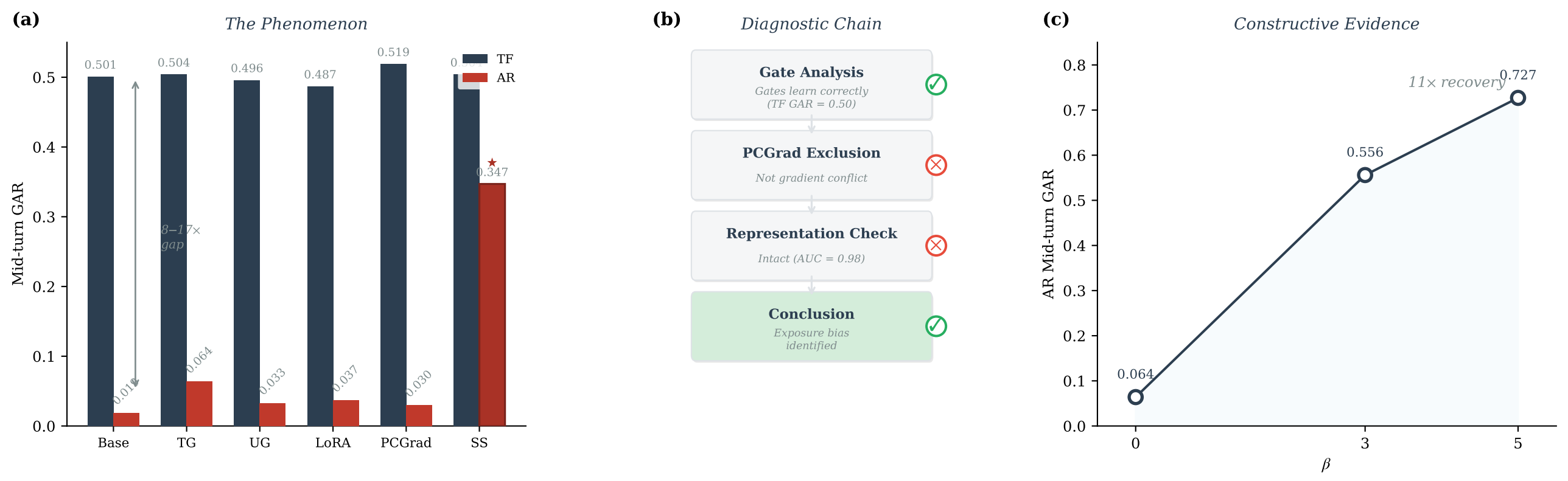
(a)
The Phenomenon
TF
AR
Mid-turn GAR
0.0
0.1
0.2
0.3
0.4
0.5
0.501
0.504
0.496
0.487
0.519
0.347
0.018
0.064
0.033
0.037
0.030
8–17× gap
Base
TG
UG
LoRA
PCGrad
SS
(b)
Diagnostic Chain
Gate Analysis
Gates learn correctly (TF GAR = 0.50)
PCGrad Exclusion
Not gradient conflict
Representation Check
Intact (AUC = 0.98)
Conclusion
Exposure bias identified
(c)
Constructive Evidence
AR Mid-turn GAR
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.064
0.556
0.727
11× recovery
0
3
5
β